

Lihe Li

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Research Interests

My research interest is Reinforcement Learning, especially in Multi-agent Reinforcement Learning.

Education

Nanjing University, Nanjing, Jiangsu, China 2023.09 - present
M.Sc. in Computer Science and Technology
Advisor: Prof. Yang Yu

Nanjing University, Nanjing, Jiangsu, China 2019.09 - 2023.07
B.E. in Artificial Intelligence
Advisor: Prof. Yang Yu

Publications (* indicates equal contribution)

LLM-Assisted Semantically Diverse Teammate Generation for Efficient Multi-agent Coordination.

[Lihe Li](#), Lei Yuan, Pengsen Liu, Tao Jiang, Yang Yu. In *The 42nd International Conference on Machine Learning (ICML)*, 2025.

Continual Multi-Objective Reinforcement Learning via Reward Model Rehearsal.

[Lihe Li](#), Ruotong Chen, Ziqian Zhang, Zhichao Wu, Yi-Chen Li, Cong Guan, Yang Yu, Lei Yuan. In *The 33rd International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.

Learning to Coordinate with Anyone

Lei Yuan, [Lihe Li](#), Ziqian Zhang, Feng Chen, Tianyi Zhang, Cong Guan, Yang Yu, Zhi-Hua Zhou. In *The Fifth Distributed Artificial Intelligence Conference (DAI)*, **Best Paper Award**, 2023.

Multi-agent Continual Coordination via Progressive Task Contextualization

Lei Yuan, [Lihe Li](#), Ziqian Zhang, Fuxiang Zhang, Cong Guan, Yang Yu. In *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*.

A Survey of Progress on Cooperative Multi-agent Reinforcement Learning in Open Environment

Lei Yuan, Ziqian Zhang, [Lihe Li](#), Cong Guan, Yang Yu. In *arxiv:2312.01058*.

Learning to Reuse Policies in State Evolvable Environments.

Ziqian Zhang*, Bohan Yang*, [Lihe Li](#), Yuqi Bian, Ruiqi Xue, Feng Chen, Yi-Chen Li, Lei Yuan, Yang Yu. In *The 42nd International Conference on Machine Learning (ICML)*, 2025.

Efficient Multi-agent Offline Coordination via Diffusion-based Trajectory Stitching.

Lei Yuan, Yuqi Bian, [Lihe Li](#), Ziqian Zhang, Cong Guan, Yang Yu. In *The 13th International Conference on Learning Representations (ICLR)*, 2025

Multi-Agent Domain Calibration with a Handful of Offline Data.

Tao Jiang*, Lei Yuan*, [Lihe Li](#), Cong Guan, Zongzhang Zhang, Yang Yu. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*, 2024.

Robust Multi-agent Communication via Multi-view Message Certification

Lei Yuan, Tao Jiang, [Lihe Li](#), Feng Chen, Zongzhang Zhang, Yang Yu. In *Science China Information Sciences (SCIS)*.

Fast Teammate Adaptation in the Presence of Sudden Policy Change

Ziqian Zhang*, Lei Yuan*, Lihe Li, Ke Xue, Chengxing Jia, Cong Guan, Chao Qian, Yang Yu. In *Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.

Dynamics Adaptive Safe Reinforcement Learning with a Misspecified Simulator.

Ruiqi Xue, Ziqian Zhang, Lihe Li, Feng Chen, Yi-Chen Li, Yang Yu, Lei Yuan. In *Joint European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, 2024.

Cost-aware Offline Safe Meta Reinforcement Learning with Robust In-Distribution Online Task Adaptation.

Cong Guan, Ruiqi Xue, Ziqian Zhang, Lihe Li, Yi-Chen Li, Lei Yuan, Yang Yu. In *The 23rd International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, 2024.

Robust Multi-agent Coordination via Evolutionary Generation of Auxiliary Adversarial Attackers

Lei Yuan*, Ziqian Zhang*, Ke Xue, Hao Yin, Feng Chen, Cong Guan, Lihe Li, Chao Qian, Yang Yu. In *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI)*, Oral Presentation, 2023.

Efficient Human-AI Coordination via Preparatory Language-based Convention

Cong Guan, Lichao Zhang, Chunpeng Fan, Yi-Chen Li, Feng Chen, Lihe Li, Yunjia Tian, Lei Yuan, Yang Yu. In *The Twelfth International Conference on Learning Representations (ICLR)*, Workshop on Large Language Model (LLM) Agents, 2024.

Honors and Awards

National Scholarship	2024
Best Paper Award of The Fifth Distributed Artificial Intelligence Conference (DAI 2023)	2023
Outstanding Bachelor's Thesis of Nanjing University	2023
Outstanding Graduate of Nanjing University	2023
The Egret Scholarship	2022
The People's Scholarship	2020, 2021

Services

Reviewer for NeurIPS	2025
Reviewer for COLM	2025
Reviewer for ICML	2025
Reviewer for ICLR	2025
Reviewer for IEEE TETCI	2024

Teaching

Teaching Assistant of "Intro to AI", Nanjing University	2024 Fall
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Skills

- Programming Skills: Python, MATLAB, C/C++, bash, HTML, $\LaTeX$
- Language Skills: Mandarin (native), English (CET4 614, CET6 603), Cantonese (native)